

Effects of Surface Type and Drop Height on the Bounce Ratio of a Ball

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Objective of the Study

● **Study Goal:** To investigate the effects of surface type and drop height on the rebound behavior of a ball.

● **Response Variable:**

$$\text{Bounce Ratio} = \frac{\text{Bounce Height}}{\text{Drop Height}}$$

● **Factors**

- **Surface Type** (3 levels: Brick, Concrete, Blacktop)
- **Drop Height** (3 levels: 38", 50", 62")

● **Controlled variables**

- **Ball type** (Spalding Grip Control Size 7)
- **Ball Pressure** (8 psi)

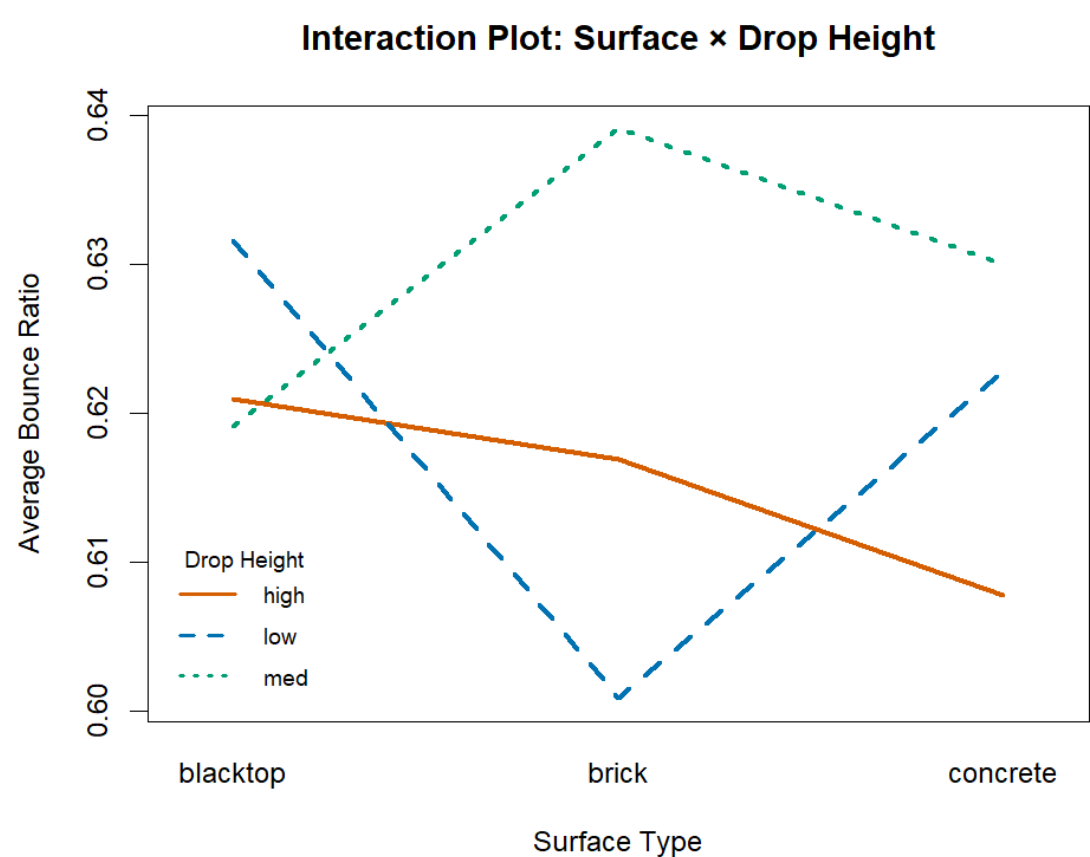
Data

The table below shows the recorded bounce heights (in inches) for each trial.

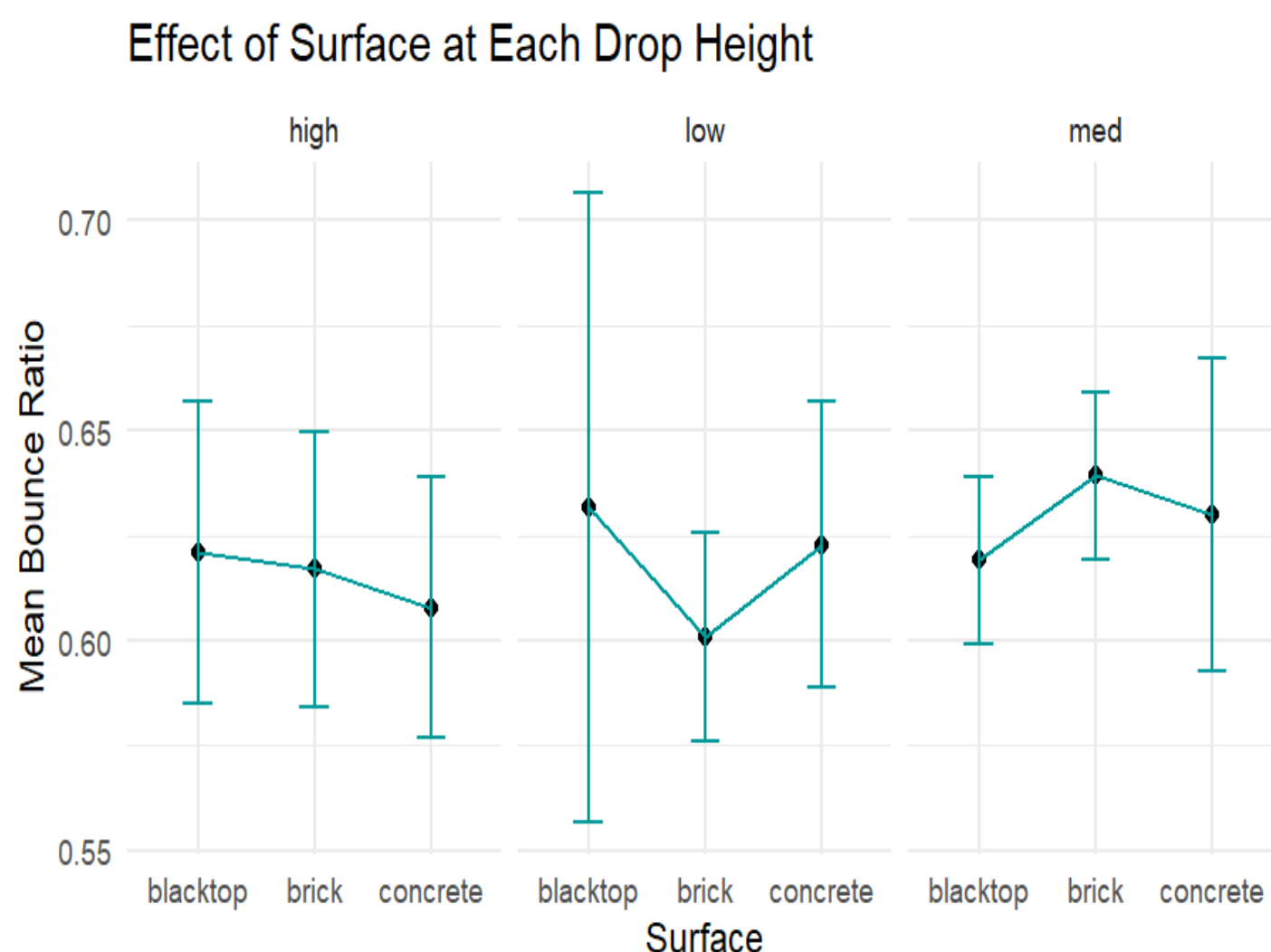
	38"	50"	62"
Brick	23.250	31.500	38.125
	22.750	32.125	39.125
	22.500	37.500	37.500
Concrete	24.250	30.750	36.800
	23.500	32.250	38.000
	23.250	31.500	38.250
Blacktop	25.250	31.250	37.500
	23.750	30.500	39.250
	23.000	31.125	38.750

Interaction Plots

This interaction plot displays evidence of interaction between drop height and surface type.



Simple Effects



Drop Height

At low drop height:

- Brick significantly lower than blacktop ($p \approx 0.059$)

At medium drop height:

- No significant differences

At high drop height:

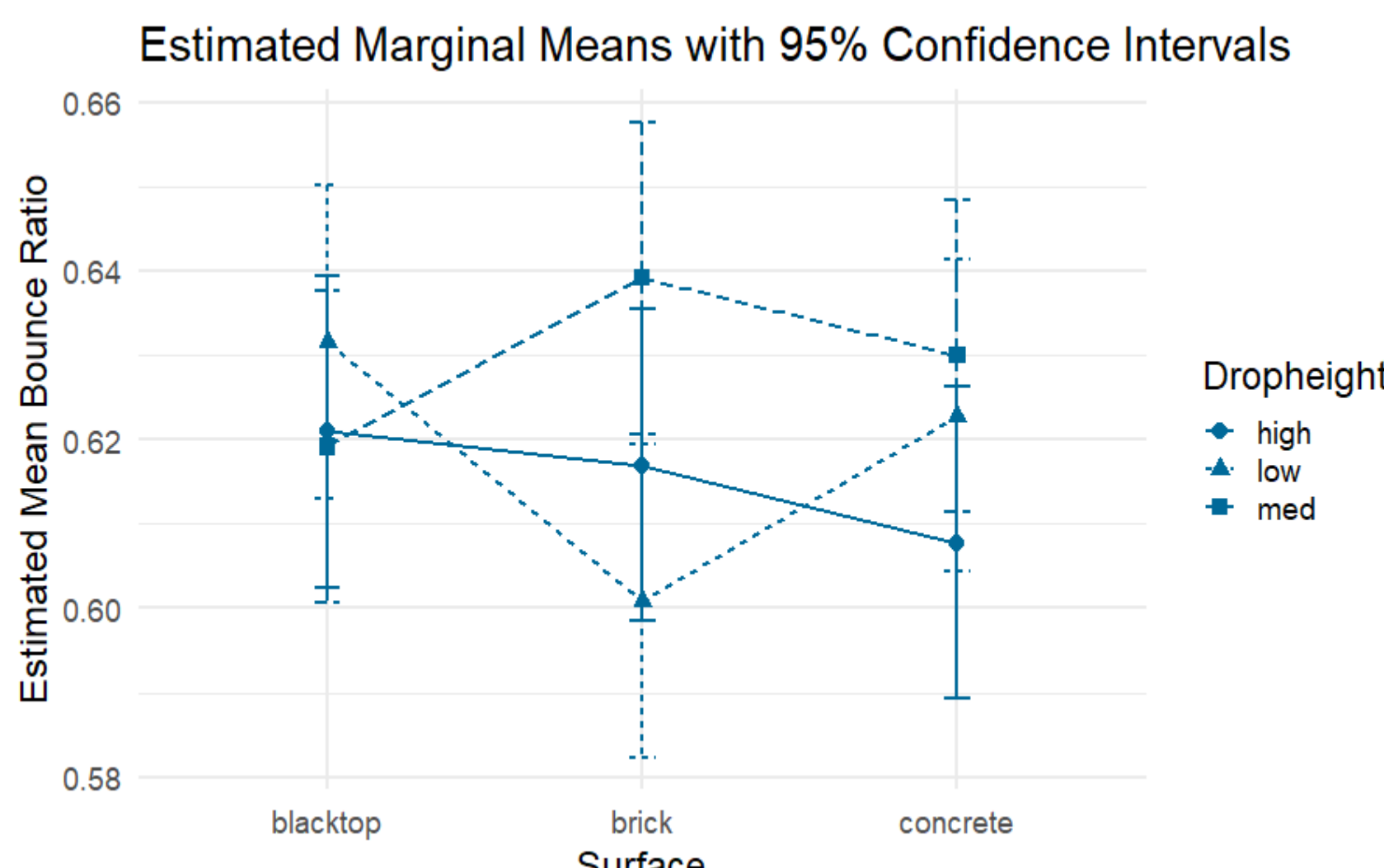
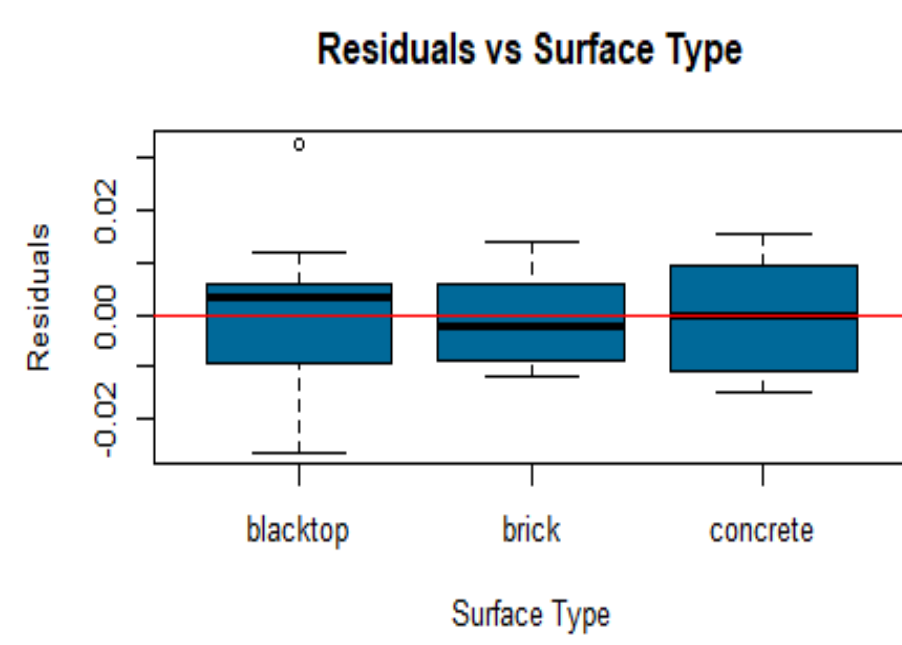
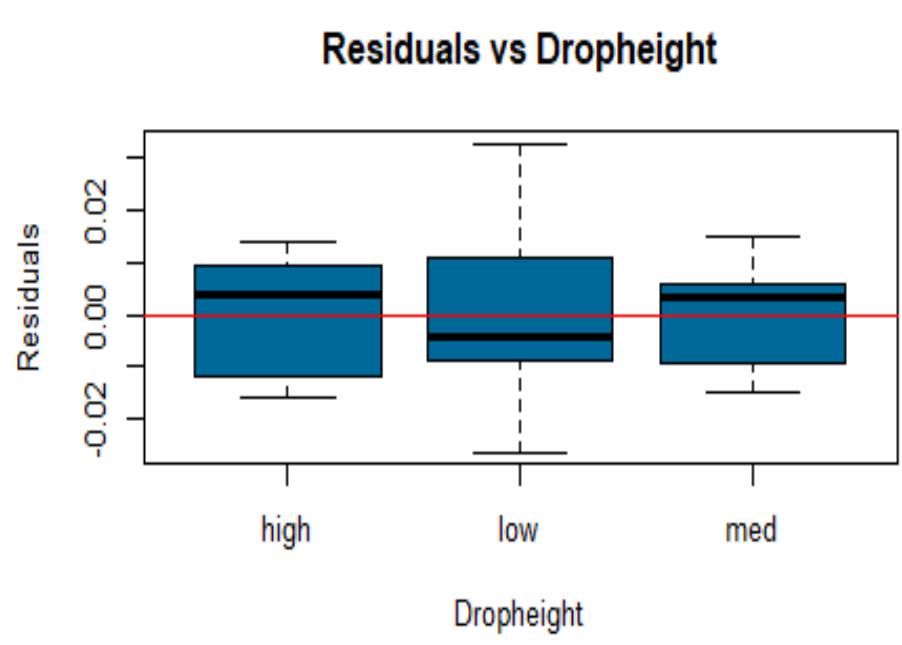
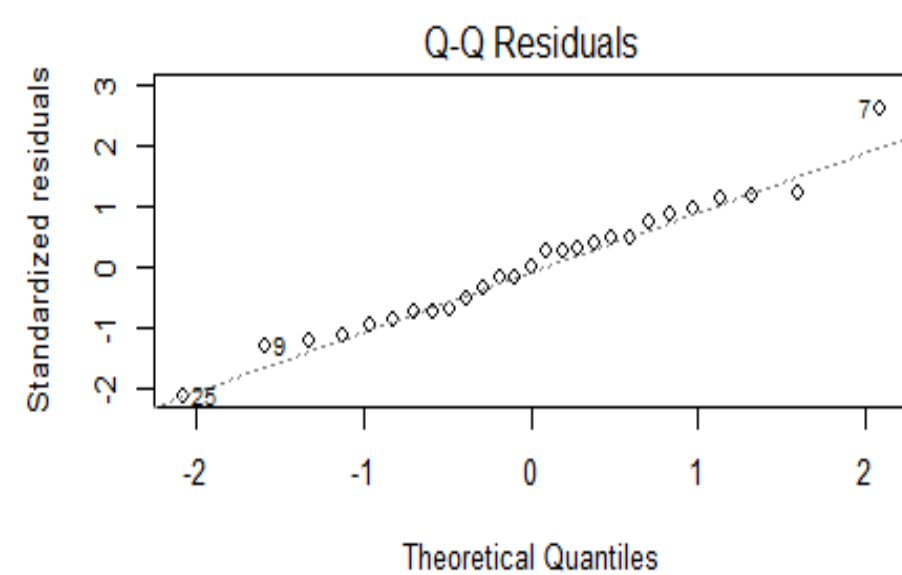
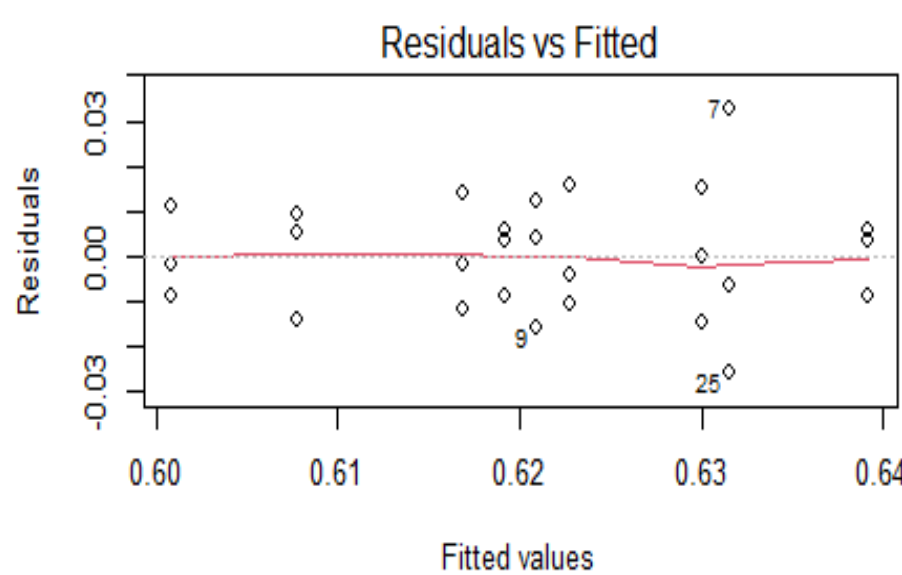
- No significant differences

Statistical Analysis

Model: A two-way ANOVA with interaction was used:

$$\text{Bounce Ratio} \sim \text{Surface} + \text{Drop Height} + \text{Surface} \times \text{Drop Height}$$

Source	Df	Sum Sq	Mean Sq	F-value	p-value
Surface	2	0.000118	0.0000589	0.253	0.7789
Drop Height	2	0.001001	0.0005005	2.151	0.1453
Surface x Drop Height	4	0.002257	0.0005643	2.425	0.0857
Residuals	18	0.004188	0.0002327		



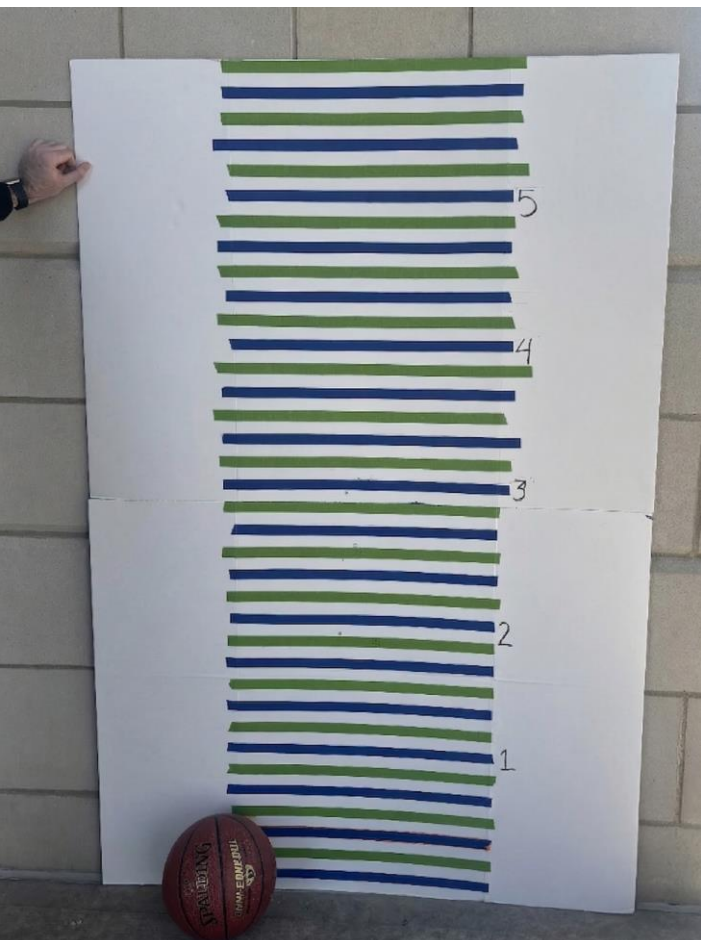
- A significant interaction between surface type and drop height was observed at $\alpha = 0.10$.
- This indicates that the effect of surface type depends on drop height, as illustrated by the Estimated Marginal Means plot.
- Residual plots indicate no major violations of ANOVA assumptions.

Key Findings

- A significant interaction between drop height and surface type was observed at $\alpha = 0.10$, consistent with the non-parallel trends seen in the interaction plot
- Bounce ratio depends on the combination of surface type and drop height
- Differences in bounce ratio across treatments are relatively small
- Results should be interpreted cautiously due to the small sample size and experimental variability

Experimental Setup

The experiment was conducted outdoors on March 30, 2026, between 10 am and 11 am. Environmental conditions included moderate wind (approximately 15 mph with gusts up to 30 mph), temperatures ranging from 61°F to 64°F, and humidity decreasing from 71% to 65%.



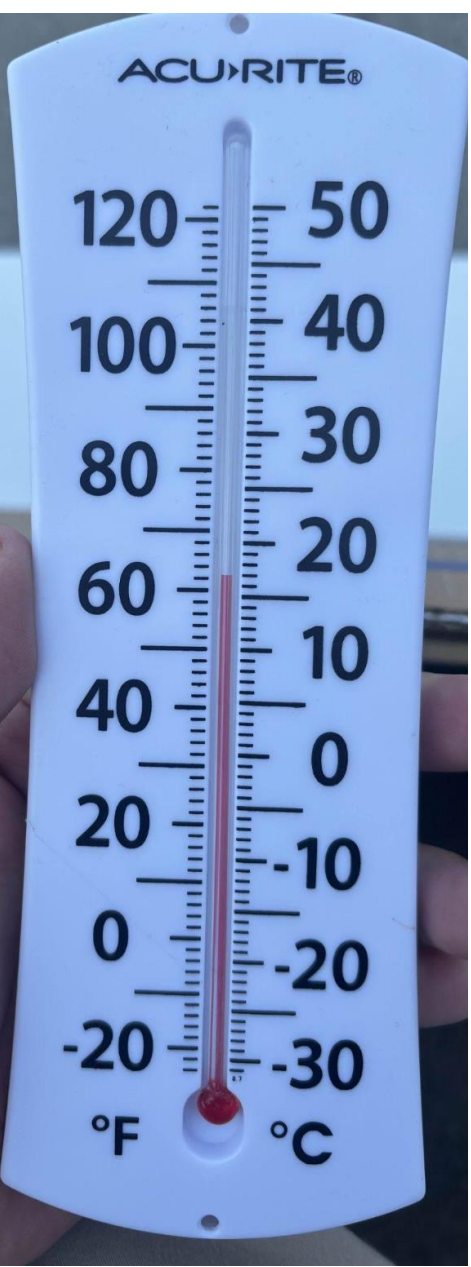
Concrete



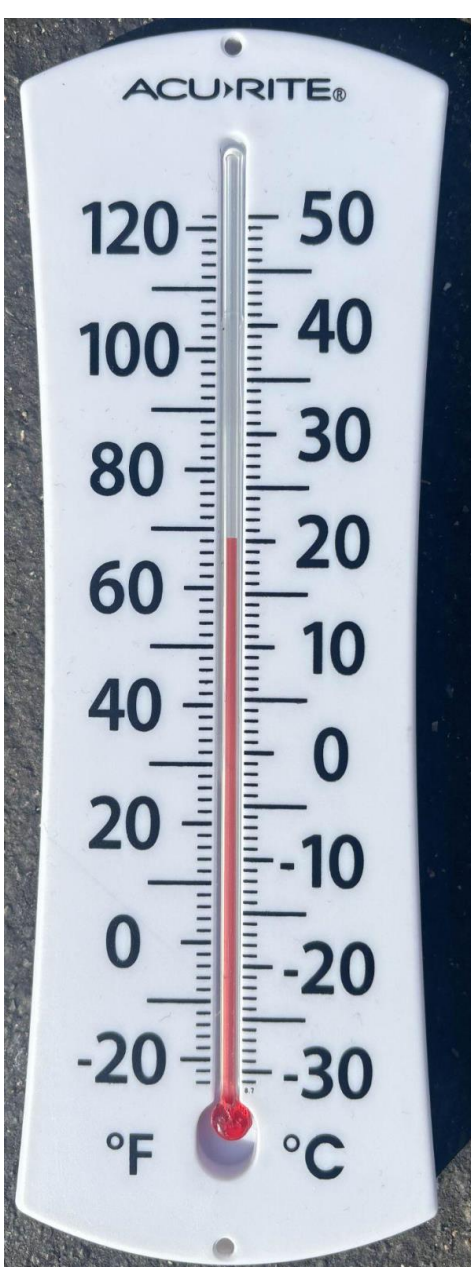
Blacktop



Brick



Before



After



Pressure

Experimental Design:

- Factorial design (3×3)
- Number of replicates ($n = 3$ per treatment)
- Total runs = 27

Discussion & Conclusions

The results of this experiment indicate a statistically significant interaction between surface type and drop height at $\alpha = 0.10$. This suggests that the effect of surface type depends on the level of drop height.

Key Findings:

- A significant interaction was observed, consistent with the non-parallel trends in the interaction plot
- Bounce ratio depends on the combination of surface type and drop height
- Differences in bounce ratio across treatments are relatively small

These results should be interpreted with caution due to several limitations:

- Small sample size (only three replicates per treatment)
- Uncontrolled outdoor conditions (wind, temperature, humidity)
- Potential variability due to manual measurement and experimental setup

Future studies should increase the number of replicates and conduct the experiment in a more controlled environment.

Acknowledgement

This work was completed as part of MTH 547 Design of Experiments under the advisement of Dr. Ying-Ju Chen.